

Southeast University

Lab Report 03

JavaScript Functions and DOM Interaction

Course Title: Web and Internet Programming Lab

Course Code: CSE472



Submitted to

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1. Objectives and project overview

Lab 3 extends the event page with JavaScript functions that respond to button clicks. It displays registration status, checks available seats, greets a student by name, and shows a workshop reminder.

Objectives

- Connect an external JavaScript file to an HTML page.
- Use functions and onclick handlers to respond to user actions.
- Select elements with `getElementById()`, read `input.value`, and update `textContent`.
- Use a variable and an if/else condition to display seat availability.

Tools and source files

Check / concept	Observation
index.html	Event content, four interaction buttons and feedback elements.
js/script.js	Seat variable and four functions.
css/style.css	Small-screen rules for images, tables and inputs.
assets/images/event picture.jpg	Local event photograph.

Google Chrome displays the application. Playwright and Node.js were used for automated checks, screenshots and PDF preparation.

Implementation scope

The supplied script sets `availableSeats` to 0. Status and reminder text are static messages. The form does not save a registration or change the seat count.

2. JavaScript source code

```
// Restore the demonstration seat count to 12 after testing the full-workshop case.
let availableSeats = 0;

function checkRegistration() {
    let message = document.getElementById("registrationStatus");
    message.textContent = "Registration is currently open.";
}

function checkSeats() {
    let message = document.getElementById("seatMessage");
    if (availableSeats > 0) {
        message.textContent = "Seats are available. Remaining seats: " +
availableSeats;
    } else {
        message.textContent = "Sorry, no seats are available.";
    }
}

function showGreeting() {
    let name = document.getElementById("studentName").value;
    let output = document.getElementById("greetingMessage");
    output.textContent = "Welcome, " + name + "!";
}

function showReminder() {
    let message = document.getElementById("reminderMessage");
    message.textContent = "Please bring your laptop and arrive ready to practise HTML,
CSS and JavaScript.";
}
```

Complete supplied js/script.js. The comment suggests restoring 12 seats after testing, but the current executable value is 0; this report documents that state.

3. Event handling and DOM concepts

Check / concept	Observation
onclick	Each type="button" control calls a named function without submitting the form.
getElementById()	Finds the paragraph or input whose ID matches the supplied string.
.value	Reads the current full-name input as a string.
textContent	Replaces output text; HTML-like characters are displayed literally.
if / else	availableSeats > 0 selects the positive-seat message; zero selects the unavailable message.
aria-live="polite"	Marks feedback regions so assistive technology can announce updates.

Button and feedback connection

```
<button type="button" onclick="checkSeats()">
  Check Seat Availability
</button>
<p id="seatMessage" aria-live="polite">
  Seat information will appear here.
</p>
```

Responsive CSS retained in this lab

```
/* No Lab 02 stylesheet was supplied. Keep the original HTML appearance
   and add only small-screen overflow fixes for the Lab 03 copy. */
img {
  max-width: 100%;
  height: auto;
}

table {
  max-width: 100%;
  overflow-wrap: anywhere;
}

input {
  max-width: 100%;
  box-sizing: border-box;
}
```

4. Browser output — interactive messages

Workshop Registration Checks

Check Registration Status

Registration is currently open.

Check Seat Availability

Sorry, no seats are available.

Show Workshop Reminder

Please bring your laptop and arrive ready to practise HTML, CSS and JavaScript.

Figure 1. Actual outputs with the supplied availableSeats = 0 value. Registration is open while no seats are available; these messages are independent.

Full Name:

Mst. Mahbuba Yasmin M Show Greeting

Welcome, Mst. Mahbuba Yasmin Maria!

Figure 2. The greeting function reads the supplied student name and updates its feedback paragraph.

Workshop Registration Checks

Check Registration Status

Registration is currently open.

Check Seat Availability

Seats are available. Remaining seats: 12

Show Workshop Reminder

Please bring your laptop and arrive ready to practise HTML, CSS and JavaScript.

Figure 3. Positive branch tested with availableSeats = 12 in browser memory only. No source edit was made.

5. Verification, limitations and conclusion

- PASS: Event image loads, all five navigation anchors have targets, form labels match inputs, and the schedule has four activity rows.
- PASS: Registration status, zero-seat feedback and workshop reminder match the script.
- OBSERVED: An empty name produces “Welcome, !”; this function does not validate or trim the input.
- PASS: Setting availableSeats to 12 in the browser exercises the positive branch. The source file remains at 0.
- PASS: The page fits a 390px viewport without horizontal overflow.

Result

All four buttons update the intended page elements. Both branches of the seat condition were verified, and the page fits the tested mobile viewport.

Limitations and possible improvements

The greeting accepts an empty name. Adding trimming and a presence check would give clearer feedback. Registration status is not connected to available seats, and the seat count is a demonstration variable rather than live capacity. No JSON or persistent storage is implemented in this stage.

Conclusion

The lab demonstrates an event-driven sequence: a click invokes a function, the function reads or evaluates data, and the DOM displays the result. These functions provide a foundation for later form validation and browser storage.

Source and preparation note

Source: the supplied lab-03-javascript folder. Repository: <https://github.com/Mariya10040/Web-Internet-Programming-Lab>.

Lab source: <https://github.com/Mariya10040/Web-Internet-Programming-Lab/tree/main/lab-03-javascript>.